

ChannelAttribution

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Markov Model for Online Multi-Channel Attribution Advertisers use a variety of online marketing channels to reach consumers and they want to know the degree each channel contributes to their marketing success. This is called online multichannel attribution problem. In many cases, advertisers approach this problem through some simple heuristics methods that do not take into account any customer interactions and often tend to underestimate the importance of small channels in marketing contribution. This package provides a function that approaches the attribution problem in a probabilistic way. It uses a k-order Markov representation to identify structural correlations in the customer journey data. This would allow advertisers to give a more reliable assessment of the marketing contribution of each channel. The approach basically follows the one presented in Eva Anderl, Ingo Becker, Florian v. Wangenheim, Jan H. Schumann (2014). Differently from them, we solved the estimation process using stochastic simulations. In this way it is also possible to take into account conversion values and their variability in the computation of the channel importance. The package also contains a function that estimates three heuristic models (first-touch, last-touch and linear-touch approach) for the same problem.

```
ChannelAttribution.auto_markov_model ( Data, var_path, var_conv, var_null, var_value=None,
max_order=10, roc_npt=100, plot=False, nsim_start=100000.0, max_step=None, out_more=False, sep='>',
ncore=1, nfold=10, seed=0, conv_par=0.05, rate_step_sim=1.5, verbose=True, flg_adv=True )
```

Parameters

- Data** : *DataFrame*
customer journeys.
- var_path**: *string*
column of Data containing paths.
- var_conv** : *string*
column of Data containing total conversions for each path.
- var_null** : *string*
column of Data containing total paths that do not lead to conversion.
- var_value** : *string, optional, default None*
column of Data containing revenue for each path
- max_order** : *int, default 10*
maximum Markov Model order to be considered.
- roc_npt**: *int, default 100*
number of points to be used for the approximation of roc curve.
- plot**: *bool, default True*
if True, a plot with penalized auc with respect to order will be displayed.

nsim_start : *int, default 1e5*
minimum number of simulations to be used in computation.

max_step : *int, default None*
maximum number of steps for a single simulated path. if NULL, it is the maximum number of steps found into Data.

out_more : *bool, default False*
if True, transition probabilities between channels and removal effects will be returned.

sep : *string, default ">"*
separator between the channels.

ncore : *int, default 1*
number of threads to be used in computation.

nfold : *int, default 10*
how many repetitions to be used to verify if convergence has been reached at each iteration.

seed : *int, default 0*
random seed. Giving this parameter the same value over different runs guarantees that results will not vary.

conv_par : *double, default 0.05*
convergence parameter for the algorithm. The estimation process ends when the percentage of variation of the results over different repetitions is less than convergence parameter.

rate_step_sim : *double, default 0*
number of simulations used at each iteration is equal to the number of simulations used at previous iteration multiplied by rate_step_sim.

verbose : *bool, default True*
if True, additional information about process convergence will be shown.

flg_adv : *bool, default True*
if True, ChannelAttribution Pro banner is printed.

Returns

list of DataFrames

result: DataFrame

(column) channel_name : channel names (column) total_conversions : conversions attributed to each channel (column) total_conversion_value : revenues attributed to each channel

transition_matrix : DataFrame

(column) channel_from: channel from (column) channel_to : channel to (column) transition_probability : transition probability from channel_from to channel_to

removal_effects:

(column) channel_name : channel names (column) removal_effects_conversion : removal effects for each channel calculated using total conversions (column) removal_effects_conversion_value : removal effects for each channel calculated using revenues

Examples

Load Data

```
>>> import pandas as pd
>>> from ChannelAttribution import *
>>> Data = pd.read_csv('https://channelattribution.io/csv/Data.csv', sep=";")
```

Estimate an automatic Markov model

```
>>> auto_markov_model(Data, "path", "total_conversions", "total_null")
```

ChannelAttribution.**choose_order** (*Data*, *var_path*, *var_conv*, *var_null*, *max_order*=10, *sep*= '>', *ncore*=1, *roc_npt*=100, *plot*=True, *flg_adv*=True)

Find the minimum Markov Model order that gives a good representation of customers' behaviour for data considered. It requires paths that do not lead to conversion as input. Minimum order is found maximizing a penalized area under ROC curve.

Parameters

Data : *DataFrame*

customer journeys.

var_path: *string*

column of Data containing paths.

var_conv : *string*

column of Data containing total conversions for each path.

var_null : *string*

column of Data containing total paths that do not lead to conversion.

max_order : *int*, *default* 10

maximum Markov Model order to be considered.

sep : *string*, *default* ">"

separator between the channels.

ncore : *int*, *default* 1

number of threads to be used in computation.

roc_npt: *int*, *default* 100

number of points to be used for the approximation of roc curve.

plot: *bool*, *default* True

if True, a plot with penalized auc with respect to order will be displayed.

flg_adv : *bool*, *default* True

if True, ChannelAttribution Pro banner is printed.

Returns list

roc : *list DataFrame one for each order considered*

(column) tpr: true positive rate (column) fpr: false positive rate

auc : *DataFrame with the following columns*

(column) order: markov model order (column) auc: area under the curve

(column) pauc: penalized auc

suggested_order : *int*

estimated best order

Examples

Estimate best markov model order for your data

Load Data

```
>>> import pandas as pd
>>> from ChannelAttribution import *
>>> Data = pd.read_csv('https://channelattribution.io/csv/Data.csv', sep=";")
```

```
>>> choose_order(Data, var_path="path", var_conv="total_conversions",
var_null="total_null")
```

ChannelAttribution.**heuristic_models** (*Data*, *var_path*, *var_conv*, *var_value*=None, *sep*= '>', *flg_adv*=True)

Estimate three heuristic models (first-touch, last-touch and linear) from customer journey data.

Parameters

Data : *DataFrame*
customer journeys.

var_path: *string*
column of Data containing paths.

var_conv : *string*
column of Data containing total conversions for each path.

var_value : *string, optional, default None*
column of Data containing revenue for each path.

sep : *string, default ">"*
separator between the channels.

flg_adv : *bool, default True*
if True, ChannelAttribution Pro banner is printed.

Returns**DataFrame**

(column) *channel_name* : channel names (column) *first_touch_conversions* : conversions attributed to each channel using first touch attribution. (column) *first_touch_value* : revenues attributed to each channel using first touch attribution. (column) *last_touch_conversions* : conversions attributed to each channel using last touch attribution. (column) *last_touch_value* : revenues attributed to each channel using last touch attribution. (column) *linear_touch_conversions* : conversions attributed to each channel using linear attribution. (column) *linear_touch_value* : revenues attributed to each channel using linear attribution.

Examples**Load Data**

```
>>> import pandas as pd
>>> from ChannelAttribution import *
>>> Data = pd.read_csv('https://channelattribution.io/csv/Data.csv', sep=";")
```

Estimate heuristic models on total conversions

```
>>> heuristic_models(Data, "path", "total_conversions")
```

Estimate heuristic models on total conversions and total revenues

```
>>> heuristic_models(Data, "path", "total_conversions", \
>>> var_value="total_conversion_value")
```

`ChannelAttribution.install_channelattributionpro ( token: str | None = None )`

Install ChannelAttribution Pro (binary wheel) for the current environment.

This installer detects your OS, architecture, and Python version, requests a prebuilt wheel (or triggers a build) from ChannelAttribution Pro's build service via **HTTPS POST**, resolves the final package URL, and installs it with *pip*. Whether the install succeeds or fails, it sends **one final** notification to *build_check_email.php* with a compact info payload.

Parameters

token : *str, optional*
Access token for the build service. If omitted, the function reads the environment variable *CHPRO_TOKEN*. If neither is provided, the installer prints a message and returns. If the token is invalid or expired, the installer prints **"Token non valid or expired. Write to info@channelattribution.io."** and returns.

Returns**None**

The function performs installation as a side effect and writes progress to stdout. On error, it prints diagnostic information and returns without raising exceptions.

Notes

- Depending on load, the remote build step may take several minutes.

- On macOS, *gcc* typically maps to Clang; compiler info is reported accordingly.
- The Linux baseline targets ManyLinux2014 for broad compatibility.
- The helper auto-installs *requests* (and *distro* on Linux) into the **current interpreter**.

Examples

Basic usage with explicit token

```
>>> from ChannelAttribution import install_channelattributionpro
>>> install_channelattributionpro(token="YOUR_TOKEN_HERE")
```

After installation

```
>>> import ChannelAttributionPro
```

`ChannelAttribution.markov_model ( Data, var_path, var_conv, var_value=None, var_null=None, order=1, nsim_start=100000.0, max_step=None, out_more=False, sep='>', ncore=1, nfold=10, seed=0, conv_par=0.05, rate_step_sim=1.5, verbose=True, flg_adv=True )`

Estimate a k-order Markov model from customer journey data. Differently from `markov_model`, this function iterates estimation until a desired convergence is reached and enables multiprocessing.

Parameters

Data : *DataFrame*

customer journeys.

var_path: *string*

column of Data containing paths.

var_conv : *string*

column of Data containing total conversions for each path.

var_value : *string, optional, default None*

column of Data containing revenue for each path.

var_null : *string*

column of Data containing total paths that do not lead to conversion.

order : *int, default 1*

Markov model order.

nsim_start : *int, default 1e5*

minimum number of simulations to be used in computation.

max_step : *int, default None*

maximum number of steps for a single simulated path. if NULL, it is the maximum number of steps found into Data.

out_more : *bool, default False*

if True, transition probabilities between channels and removal effects will be returned.

sep : *string, default ">"*

separator between the channels.

ncore : *int, default 1*

number of threads to be used in computation.

nfold : *int, default 10*

how many repetitions to be used to verify if convergence has been reached at each iteration.

seed : *int, default 0*

random seed. Giving this parameter the same value over different runs guarantees that results will not vary.

conv_par : *double, default 0.05*

convergence parameter for the algorithm. The estimation process ends when the percentage of variation of the results over different repetitions is less than convergence parameter.

rate_step_sim : *double, default 0*

number of simulations used at each iteration is equal to the number of simulations used at previous iteration multiplied by rate_step_sim.

verbose : *bool, default True*

if True, additional information about process convergence will be shown.

fig_adv : *bool, default True*

if True, ChannelAttribution Pro banner is printed.

Returns

list of DataFrames

result: DataFrame

(column) channel_name : channel names (column) total_conversions : conversions attributed to each channel (column) total_conversion_value : revenues attributed to each channel

transition_matrix : DataFrame

(column) channel_from: channel from (column) channel_to : channel to (column) transition_probability : transition probability from channel_from to channel_to

removal_effects:

(column) channel_name : channel names (column) removal_effects_conversion : removal effects for each channel calculated using total conversions (column) removal_effects_conversion_value : removal effects for each channel calculated using revenues

Examples

Load Data

```
>>> import pandas as pd
>>> from ChannelAttribution import *
>>> Data = pd.read_csv('https://channelattribution.io/csv/Data.csv', sep=";")
```

Estimate a Makov model using total conversions

```
>>> markov_model(Data, "path", "total_conversions")
```

Estimate a Makov model using total conversions and revenues

```
>>> markov_model(Data, "path", "total_conversions",
var_value="total_conversion_value")
```

Estimate a Makov model using total conversions, revenues and paths that do not lead to conversions

```
>>> markov_model(Data, "path", "total_conversions",
var_value="total_conversion_value", var_null="total_null")
```

Estimate a Makov model returning transition matrix and removal effects

```
>>> markov_model(Data, "path", "total_conversions",
var_value="total_conversion_value", var_null="total_null", out_more=True)
```

Estimate a Markov model using 4 threads

```
>>> markov_model(Data, "path", "total_conversions",
var_value="total_conversion_value", ncore=4)
```

`ChannelAttribution.request_token_channelattributionpro ( email: str, endpoint: str = 'https://app.channelattribution.io/genpkg/generate_token.php', timeout: int = 10, verify_ssl: bool = True ) → str`
 Send an email address to ChannelAttribution Pro's `generate_token.php` endpoint and return the server's response body.

Parameters

email : *str*

Target email address to which the token should be sent. Must be **non-empty**. (No syntactic validation is performed by this function.)

endpoint : *str*, default `"https://app.channelattribution.io/genpkg/generate_token.php"`

Full URL of the token-generation PHP endpoint. You can override this for testing.

timeout : *int*, default `10`

Timeout in seconds applied to the HTTP request.

verify_ssl : *bool*, default `True`

Whether to verify the server's TLS certificate. Set to `False` only in controlled testing environments.

Returns **str** The exact response body (trimmed) returned by the server. Typical values include: - "We've sent the token to your email address. ..." - "Token already generated" - "Provider not admitted" - "mail not valid" - "db query error", "db connection error", etc.

Raises

ValueError

If `email` is empty.

RuntimeError

For network/TLS issues (connection errors, DNS failure, timeouts, SSL problems), with a message prefixed by `"network_or_ssl_error:"` or `"request_error:"`.

Notes

- To validate the email syntax before calling this function, do it in the caller (e.g., from `email.utils import parseaddr` and check that `parseaddr(email)[1]` is non-empty).
- Honors system proxy settings if they are configured for Python/OS.

Examples

Basic usage

```
>>> from ChannelAttribution import request_token_channelattributionpro
>>> request_token_channelattributionpro("john.black@company.com")
```

`ChannelAttribution.transition_matrix ( Data, var_path, var_conv, var_null, order=1, sep='>', flg_equal=True, flg_adv=True )`

Estimate a k-order transition matrix from customer journey data.

Parameters

Data : *DataFrame*

customer journeys.

var_path: *string*

column of Data containing paths.

var_conv : *string*

column of Data containing total conversions for each path.

var_null : *string*

column of Data containing total paths that do not lead to conversion.

order : *int*, default `1`

Markov model order.

sep : *string*, default ">"

separator between the channels.

flg_equal: *bool*, default **True**

if True, transitions from a channel to itself will be considered.

flg_adv : *bool*, default *True*

if True, ChannelAttribution Pro banner is printed.

Returns

list of DataFrames

channels: *Dataframe*

(column) id_channel : channel ids (column) channel_name : channel names

transition_matrix : *DataFrame*

(column) channel_from: id channel from (column) channel_to : id channel to

(column) transition_probability : transition probability from channel_from to channel_to

Examples

Load Data

```
>>> import pandas as pd
>>> from ChannelAttribution import *
>>> Data = pd.read_csv('https://channelattribution.io/csv/Data.csv', sep=";")
```

Estimate a second-order transition matrix using total conversions and paths that do not lead to conversion

```
>>> transition_matrix(Data, "path", "total_conversions",
    var_null="total_null", order=2)
```

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